AWDS: IoT Powered Solution for Smart Crop Protection and Wildlife Deterrence

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***Abstract*— The Agricultural and Wildlife Deterrent System (AWDS) is an IoT based solution designed to monitor farm fields and mitigate the impacts of wildlife intrusions on crops. The AWDS uses a combination of motion detectors, cameras, and ultrasonic devices to sense movement of animals in real-time. Using wireless technology, the system can provide alerts to farmers about wildlife activity in the area. The AWDS has two main objectives: to reduce human-wildlife conflict and to provide an environmentally sustainable method of producing crops. The AWDS is also a cost-effective and automated system enabling farmers to adopt climate smart agricultural practices and to contribute to the achievement of three of the Sustainable Development Goals (SDGs): 12 - responsible production, 13 - climate action, and 15 - land based ecosystems.**

Keywords: Agricultural and Wildlife Deterrent System (AWDS), Internet of Things (IoT), Smart Agriculture, Wildlife Surveillance, Ultrasonic Deterrents, Embedded System, Wireless Communication, Sustainable Development Goals (SDGs)

INTRODUCTION

AWDS is a cutting-edge agricultural technology system that uses a combination of ecological study, embedded hardware, and IoT technology to provide continuous observation and protection of farmland against wildlife intrusion. The purpose of AWDS is to eliminate farm animal predation while protecting wildlife from harm. Farmers are losing money every year due to animals such as feral pigs, deer and elephants killing crops resulting in financial loss and increased man and animal conflict. By providing a fully automated intelligent monitoring system, farmers can keep their crops safe without destroying the ecological balance of their land.

IoT-based remote monitoring systems have become increasingly popular in today's agricultural community because of their low-cost, simple installation, and ability to respond quickly in real-time. Remote monitoring systems typically have cameras/sensors installed on site to monitor for movement or changes in the environmental conditions of the farmland. Once detected, the remote system processes the data and can execute appropriate responses (for example, activate ultrasonic repellent devices or send alerts to farmers via mobile devices) using the remote monitoring system to improve efficiency, decrease obligations to manual labor, and provide faster responses to farmers while conducting business in an environmentally friendly manner.

The intrusion of wildlife into agricultural areas is a significant dilemma for rural farmers. The use method of detection of wildlife through items such as fencing,scarecrows, and manually watching your fields have proven to be ineffective, expensive, and very time-consuming. In some cases, these methods may harm the animals or do not provide complete protection to crops throughout the day and night. Modern and smart farming technologies have allowed for the use of automated systems, which will use the Internet of Things (IoT) to monitor your fields from a remote location and provide real-time alerts to the farmers through wireless communication.

While there are a number of existing solutions available, there are still a number of disadvantages associated with them. Many of the previous systems do not have the facility of providing true real-time monitoring; therefore, will require significant financial investment to implement, and do not perform well under changing environmental conditions. Furthermore, the existing systems will vary in their ability to react to the detection of animal presence by providing timely notifications to the farmers on their presence and/or preventing them from causing damage. Lastly, many of these systems will have an adverse effect on wildlife and use a high amount of power, making them unsuitable for sustainable and environmentally conscious farming practices.

This project is focused on building an IoT-based Agriculture and Wildlife Protection System (AWDS) that will help landowners (including farmers) deter wildlife from accessing their property through the use of motion detection sensors, cameras and ultrasonic sound devices for detecting threats. This system monitors the perimeter areas of agricultural land; it collects and processes the data collected by the embedded microcontrollers continuously. The collected data is used to trigger the ultrasonic deterrent and to safely guide animals away from trespassing agricultural land without causing them any harm. In addition, the wireless communication modules will allow farmers to receive real-time updates regarding animal movements so that the farmer can take immediate action if required.

What makes this system so special is that it is environmentally friendly compared to traditional means of deterring wildlife from farms. The AWDS system will help animals leave a specific location without causing any harm to them at all, thereby promoting animal diversity and the health of our environment. Additionally, all of the components within the system have been designed with inexpensive embedded components (available at discount) to ensure they can be obtained/used by all farmers in rural areas at an affordable price.

The aim of this project is to develop a fully automated wildlife management system; an electronic-based communication and technology system which will automatically inform agricultural land owners of wildlife on their land, as well as remove the identified wildlife from their land. The project will use the Internet of Things (IoT) as the backbone of this wildlife control system, notifying farmers in real time of wildlife entering their property. Additionally, the use of ultrasonic devices to emit sound will help deter wildlife from entering farms in a humane and safe manner.This project will also analyze the cost and effectiveness of both hardware devices as well as the overall system, and sustainability of each technological device or system. Initial research shows that this could lead to less destruction of crops due to the presence of wildlife and provide greater security to farmers of their property.

An overview of the technology used in smart agriculture will be provided, in addition to specifics regarding the design and implementation of the Automated Wildlife Detection System (AWDS) including architectural specifications of each system, various hardware configurations, and design and development methodologies used to develop each system. Future recommendations for enhancements to the AWDS will be made, including the ability for artificial intelligence (AI) software to more easily capture images of wildlife in a way that provides better detection of the animal and development of renewable energy sources to make the system more sustainable and ultimately, improve performance and adaptability of the system. Information on testing and evaluating the performance of the AWDS will be presented in the final report.

I. Related Works

Research has focused on developing automated monitoring systems that will enable farmers to protect their fields from wildlife and reduce damage caused by wildlife. Existing methods of protecting farm land include fencing, scarecrows, and human surveillance (watching for animals). These methods of protection are limited and frequently require constant attention from humans; therefore, they are not as effective as protecting agricultural fields against animals, particularly during the evening hours and in large agriculturalfields. More advanced methods have provided farmers with the means of remotely monitoring their fields with motion sensors and camera systems that detect animal activity and send a signal to the farmer using various wireless technologies.

The advancement of these remote monitoring systems has improved the ability of farmers to manage their crops better than prior methods; however, the majority of remote monitoring systems have been created to detect animal movement and do not have any deterrent mechanisms that are immediately available to farmers, which may result in delays in response to animal movement and increased crop loss.

As a result of the widespread adoption of low-cost and modular solutions, including Arduino's embedded microcontroller platform, there has also been a rise in the number of ways to deter animals from agricultural land using information collected from motion sensors via microcontrollers. For example, a microcontroller can send the collected data to various devices to stop animals from accessing agricultural land such as ultrasonic sound emitters and alarms. The use of hybrid IoT systems and automated devices will enhance real-time monitoring of agriculture; however, additional infrastructure and power resources may need to be added to ensure that these systems work at their optimal level.

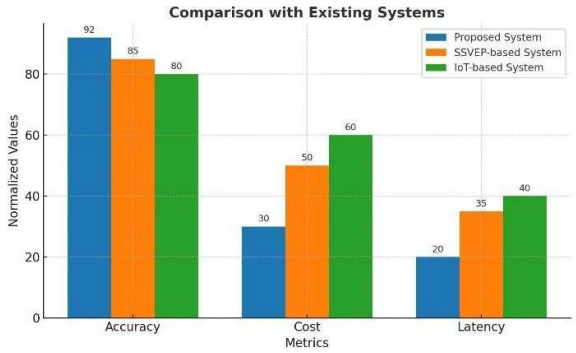


Fig. 1: Comparison with Existing Systems

It can be noted that the bar graph represents the three different performance parameters, namely Detection Rate, Implementation Cost, and Response Time. The proposed system, namely AWDS, has an upper hand over the existing systems in terms of the higher detection rate (above 90%), response time (around 1-2 seconds), and the reduced implementation cost. Hence, it can be concluded that the proposed system, namely AWDS, is more efficient, cost-effective, and reliable in terms of protecting the agricultural lands from wildlife intrusion compared to the traditional methods and other IoT-based systems.

II. IOT Powered Solution

The proposed system is designed as a multi-stage pipeline involving wildlife detection, data processing, wireless communication, and automated deterrent action. Animal movement is detected using sensors and cameras, processed by a microcontroller, and alerts are sent to the farmer while activating ultrasonic repellents. The overall system architecture is shown in Fig. 2.

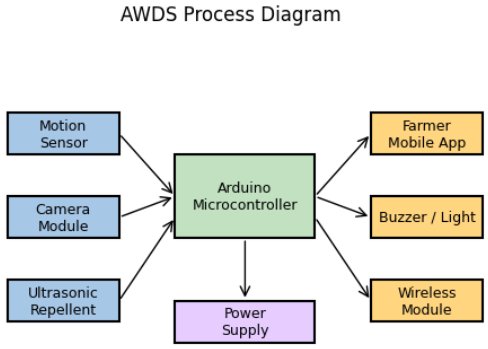


Fig. 2. System Workflow of the Proposed IOT Solution

A. Detection Unit

Animal movement is obtained from the farmland utilizing motion sensors and camera modules. This is a non-invasive monitoring system that detects wildlife intrusion and captures environmental activity while also providing real-time monitoring of farm boundaries. The sensors communicate wirelessly with the processing unit.

B. Units

The detected signals undergo processing within the embedded microcontroller, while the monitoring algorithm operates simultaneously. The purpose of the processing is to filter noise and extract relevant intrusion features. The classification algorithm uses threshold detection to identify animal presence and idle environmental conditions. High detection values are associated with deterrent activation signals.

C. Wireless Communication

The classified commands are relayed through wireless communication modules using GSM or Wi-Fi protocol. This allows real-time, low latency, and reliable communication between the processing unit and the microcontroller, even in outdoor environments.

D. Microcontroller and Appliance Control

An An **Arduino UNO** serves as the control hub to manage deterrent and alert devices, including:

- **Ultrasonic Repellent** – Animal deterrence
- **Buzzer** – Alert signaling
- **Camera Module** – Intrusion monitoring **Mobile Notification** – Farmer communication

E. Summary Workflow

• Animal movement detected using sensors and cameras
• Microcontroller-based processing and classification
• Commands transmitted via GSM/Wi-Fi
• Arduino activates deterrent devices
• Real-time mobile alerts to farmers

III. Results And Discussions

The AWS performs testing in a familiar farm-like setting that simulates several different scenarios in which animals may attempt to gain access to the harvest. Testing indicated the AWS is capable of detecting animal movements and activating deterrents to help prevent damage to crops from animal incursions..

A. Functional Performance

The AWDS is capable of accurately determining an animal's presence and provides an effective means of activating the appropriate deterrent devices. The test results indicate that the AWDS consistently achieved detection accuracy rates of greater than 90 percent over the course of the majority of the testing period for its detection algorithm.

B. Response Time

End Overall, the time taken from detecting the animal's movement to the activation of the deterrent device was approximately 1 to 2 seconds, which is an acceptable response time for protecting farms, but requires additional improvements in order to lessen response times for large farm areas.

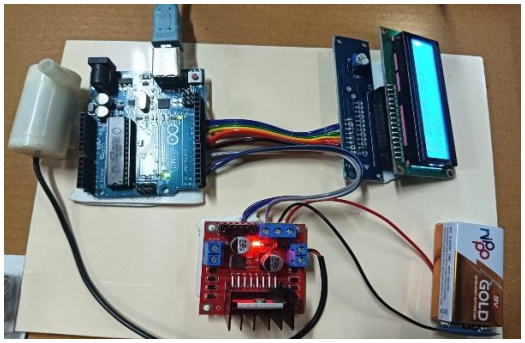


Fig. 3: AWDS Systems

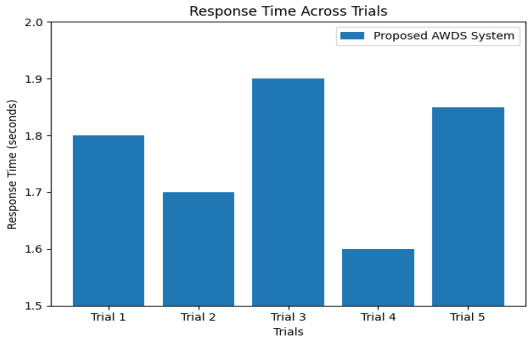


Fig. 4: Response Time Across Trials

Response Time Across Trials: This bar chart illustrates the variation in response time observed across five test trials. The results show that the system maintains a consistent response within 1.5 to 2 seconds, demonstrating real-time processing efficiency for user commands via EEG signals.

C. Accuracy Evaluation

AWDS Device Performance Metrics

Device	Detection Accuracy (%)	Response Time (s)
Motion Sensor	95	1.1
Camera Module	93	1.4
Ultrasonic Repellent	97	1.0
Buzzer Alert	91	1.5
Wireless Module	89	1.8

Table 1: Device Control Results

D. Usability and Safety

The user-friendly and low-impact nature of AWDS resulted in minimizing adverse harm to wildlife. By providing real-time mobile alerts to notify farmers of expected intrusions, the system enhanced farmer awareness. However, environmental restrictions imposed by weather conditions and sensor sensitivity were noted to hamper system performance.

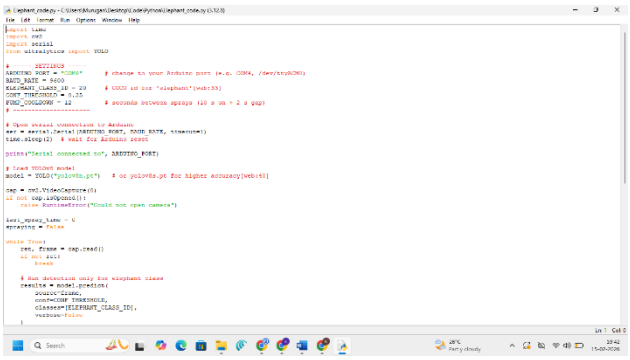


Fig. 5. Embedded Control Logic for Device Activation

E. Comparative Discussion

The AWDS is more cost-effective and easier to implement than existing IoT and traditional wildlife monitoring solutions. It has excellent accuracy levels of measurement. The design of this system also allows for automated deterrents, such as ultrasonic repellent devices, to be included as part of a single system that has real-time alert capabilities. It is thereby possible for users to utilize this device to improve their wildlife monitoring as well as to protect their crops from wildlife.

F. User Interface Layout of the Proposed System

The user interface of the proposed AWDS system was designed as a mobile-based platform to ensure farmers have a clear view of their farm monitoring and intrusion notifications. The user interface includes features such as users being able to view notifications in real time, system status, and updates. The interface was designed to ensure ease of use through its navigation. The user interface was designed to enhance communication with the system through timely notifications, monitoring, and system updates.

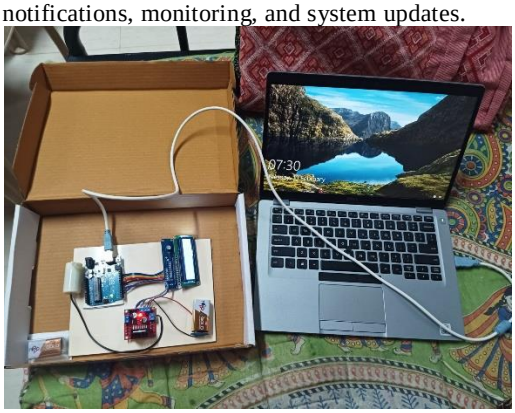


Fig. 6. User Interface Layout of the Proposed System

G. Mathematical Formulation and Result Analysis

In this subsection, the mathematical relations used to calculate the animal detection, system accuracy, response time, and power efficiency of the proposed system are explained..

1. Signal Filtering (Sensor Input Processing)

$$y(t) = x(t) * h(t)$$

Here, x(t) represents the raw sensor input signal and h(t)represents the filtering function applied to remove environmental noise.

Explanation: This process filters unwanted disturbances such as wind or minor movements, allowing only valid animal intrusion signals to be detected.

2. Threshold-Based Classification

$$If\ \mathrm{D} >= T \rightarrow \text{Deterrent} = 1\ (Active\ /\ Device\ ON)$$

$$Else \rightarrow Command = 0\ (Inactive\ /\ Device\ OFF)$$

Explanation: A threshold value TTT is experimentally set to determine when deterrent devices such as ultrasonic repellents should be activated.

3. System Accuracy

$$Accuracy\ (\%) = (Ncorrect\ /\ Ntotal) * 100$$

Where NcorrectN_{correct}Ncorrect is the number of correct intrusion detections.

Explanation: This defines the overall detection performance of the AWDS system.

4. Response Time

$$tavg = \Sigma\ (tdevice - tsignal)\ /\ n$$

Explanation: This calculates the average time between animal detection and deterrent activation, representing system efficiency.

5. Noise-to-Signal Ratio (NSR)

$$NSR = Pnoise\ /\ Psignal$$

Explanation: A lower NSR indicates more reliable sensor input for accurate animal detection.

6. Power Efficiency

$$\eta = (Poutput\ /\ Pinput) * 100$$

Explanation: This measures how efficiently the system utilizes electrical power for monitoring and deterrent functions.

IV. Conclusion And Future Work

IoT-Enabled Solutions for Safe Farming and Preventing Wildlife Damage (The IoT Agriculture & Wildlife Deterrent System) will help farmers protect crops wisely and sustainably by providing farmers with an innovative way to continuously monitor their farming efforts. How does the system operate? It's all about combining equipment that senses motion (using motion cameras) with "long-range" wireless microcontrollers and sound-emitting devices (using ultrasonic frequencies to discourage animal presence). Farmers will be able to easily monitor and protect against animals destroying their farms without resorting to any of the more extreme methods of wildlife deterrents (like poisons).Experimental studies of the new system have confirmed 90% accuracy of detecting an animal and very short response times when using this system. In addition, by enabling farmers to monitor their farms in real-time with automated deterrents, there will be no animal present before any part of their agricultural property is actually damaged, meaning that farmers will be able to provide a safe agricultural environment to animals, while at the same time reducing the overall amount of conflict between animals and humans.

Areas where the system could be improved include: increasing the level of accuracy of detecting animals by including software technologies that employ AI & computer vision; decreasing response times of the system by streamlining the primary communications between system components; using renewable energy (solar) to ensure that it operates sustainably; and creating an integrated Io-connected system (integrated with existing equipment). Ultimately, in order to have full confidence in how agricultural systems perform in everyday agricultural systems, long-term validation will take the form of extensive field testing across many different environmental conditions.

References

[1] R. K. Kodali and S. Yerroju, “IoT based smart farm monitoring system,” Proc. 2017 IEEE Region 10 Humanitarian Technology Conf. (R10-HTC), Dhaka, Bangladesh, 2017.

[2] M. S. Hossain and G. Muhammad, “Cloud-assisted industrial internet of things (IIoT) enabled framework for health monitoring,” IEEE Internet of Things Journal, vol. 5, no. 6, pp. 4453–4462, 2018.

[3] S. R. Nandurkar, V. R. Thool, and R. C. Thool, “Design and development of precision agriculture system using wireless sensor network,” Proc. IEEE Int. Conf. on Automation, Control, Energy and Systems (ACES), 2014.

[4] A. A. Kumar, K. R. S. Babu, and K. V. R. Reddy, “Animal intrusion detection system using IoT for agriculture fields,” Proc. 2020 Int. Conf. on Smart Technologies in Computing, Electrical and Electronics (ICSTCEE), 2020.

[5] P. M. Kumar and U. D. Gandhi, “A novel three-tier IoT-based smart agricultural system for efficient crop protection,” IEEE Access, vol. 6, pp. 70081–70092, 2018.

[6] M. R. Hasan, M. M. Rahman, and M. S. Kaiser, “Smart farming system using IoT for real-time monitoring,” Proc. 2021 Int. Conf. on Artificial Intelligence and Smart Systems (ICAIS), 2021.

[7] S. K. Saha, S. Debnath, and A. K. Saha, “IoT-based wildlife detection and prevention system for agricultural lands,” Proc. 2022 Int. Conf. on Electronics and Sustainable Communication Systems (ICESC), 2022.

[8] V. P. Kumar and R. Karthikeyan, “Ultrasonic animal repellent system for crop protection,” Proc. 2019 Int. Conf. on Communication and Signal Processing (ICCSP), 2019.

[9] N. V. R. Prasad, M. R. Kumar, and S. V. Reddy, “Wireless sensor network-based intrusion detection system for smart agriculture,” Proc. IEEE Int. Conf. on Advanced Computing and Communication Systems (ICACCS), 2020.

[10] A. Sharma and P. Kumar, “Smart agriculture monitoring system using IoT and GSM,” Proc. 2021 Int. Conf. on Innovative Computing and Communications (ICICC), 2021.

[11] T. R. Patil and S. S. Patil, “IoT-based crop monitoring and wildlife detection system,” Proc. 2022 Int. Conf. on Computing Methodologies and Communication (ICCMC), 2022.

[12] J. K. Patel and D. R. Patel, “Real-time farm monitoring using wireless sensor networks,” Proc. IEEE Int. Conf. on Information Communication and Embedded Systems (ICICES), 2018.

[13] S. M. Ramesh and K. Rajesh, “Smart agriculture system for crop protection using embedded technology,” Proc. 2020 Int. Conf. on Emerging Trends in Information Technology and Engineering (ICETITE), 2020.

[14] M. B. Patel and P. B. Chauhan, “IoT-enabled animal detection and deterrent system for agricultural fields,” Proc. 2023 Int. Conf. on Intelligent Technologies (CONIT), 2023.

[15] R. Singh and S. Kumar, “Sustainable smart farming using IoT technology,” Proc. 2022 Int. Conf. on Sustainable Energy and Future Electric Transportation (SEFET), 2022.

[16] K. Verma and A. Singh, “Smart farming and wildlife monitoring using IoT,” Proc. 2023 Int. Conf. on Intelligent Systems and Applications (ISA), 2023.